Order Planning Strategy Process

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This document describes an order planning strategy, which is used to consolidate all configurations across components, and trigger a wave or a stream run using those configurations.
Below are available configurations:

- Order Planning Strategy
- Selection Strategy
- Allocation Strategy
- Replenishment Strategy
- Cubing Strategy
- Consolidation Strategy
- Routing Strategy
- Picking Task Strategy
- Replenishment Task Strategy
- Pack Sort Resource Strategy
- Replenish Resource Strategy

Process Flow

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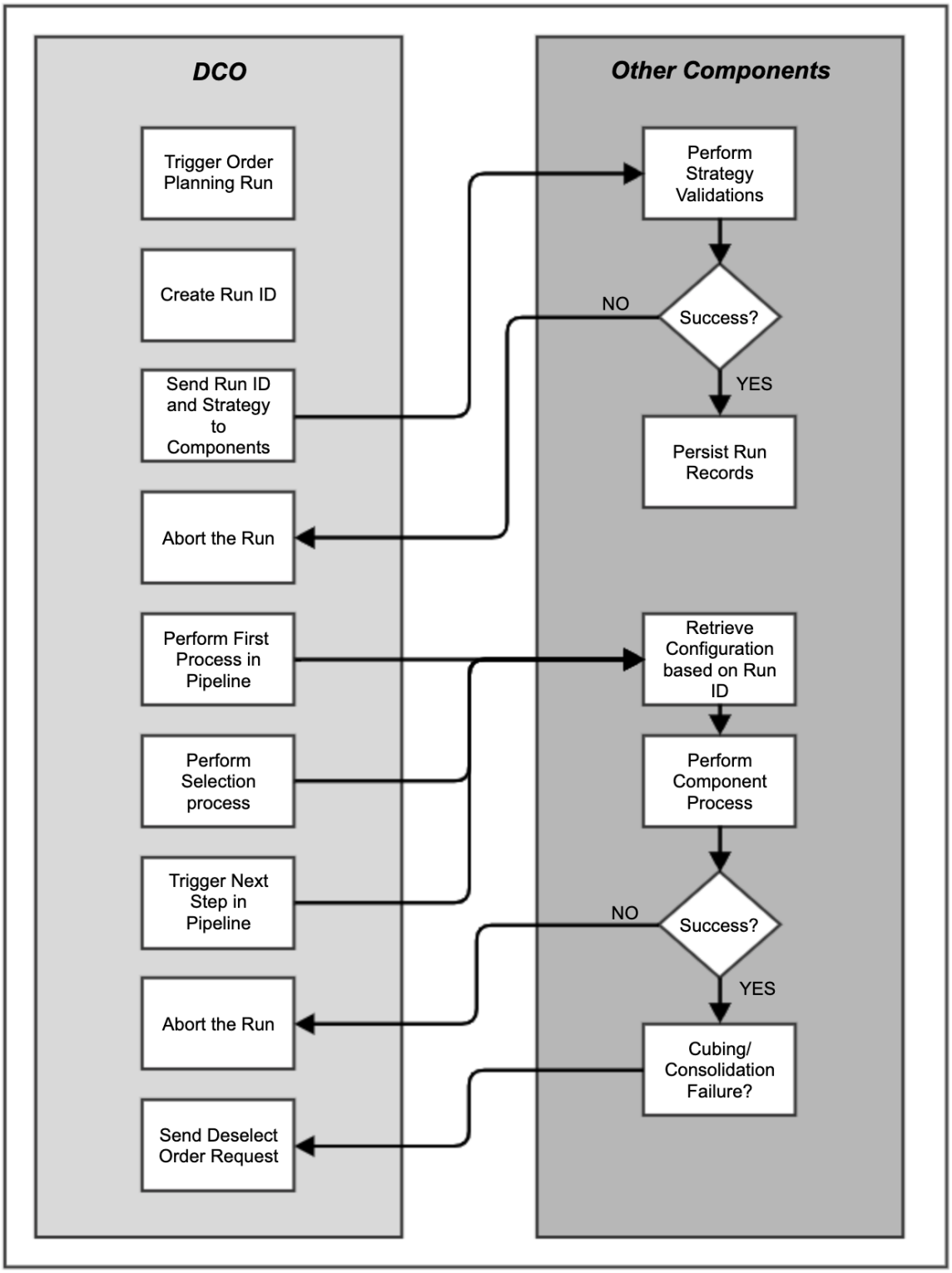

Order Planning Strategy

Order Planning Strategy (OPS) is automated according to the way the orders are selected, allocated, and released by the system to determine the best picking and shipping methods on the floor. An OPS is a unique set of strategies by which Warehouse Management processes each selected order. It allows the user to select a unique pipeline to be used which determines the sequence of steps for processing the orders. OPS provides the ability to initiate a 'Shipping Wave' or a 'Stream Run', based on the selected component strategies in the OPS. Based on the selected pipeline, OPS allows the user to select the pre-configured

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component strategies to drive order processing, thus resulting in a planning run for a wave or stream. OPS provides an option to manually initiate a planning run; alternatively, an OPS can be initiated via an automatic time-triggered scheduler.
The following activities are performed as a part of the OPS-driven planning run:

- Run ID generation:
  - DC Order selects the list of orders for the wave process based on order selection rules.
  - Sends updates to the shipment component to move the Warehouse status of shipment's pickup stop to DC Accepted for preplanned orders (Order line's shipment is not NULL).
- Order planning run ID and component strategy IDs are sent to components, which are part of the OPS pipeline.
- Each component validates the associated strategies and persists the run records.
- DCO triggers the different steps in the OPS pipeline.
- Each component performs its own process, using the configuration associated with the component strategy selected on the OPS.
- DCO orchestrates the execution of consecutive steps in the pipeline, based on response from each component in the OPS pipeline.
- DCO may abort the run or deselect an order in the run if there is any failure with 'ABORT' as the exception class in the response.
- When all the steps in the pipeline are completed, the allocations are exported to 'Work Release'.
- The run then ends successfully when all the steps in the pipeline are completed.

Strategy Configuration

The following configurations define an order planning strategy and govern the properties of an order planning/ run pipeline processing:

Configuration	Description
Allocation Strategy	The allocation strategy defines different allocation configurations that drive the allocation process, such as allocation order criteria, and allocation inventory criteria. You can refer Allocation Process Guide for more details.
Batch Consolidation	A batch consolidation configuration decides if the consolidation process has to be performed in discrete mode (consolidate each order) or batch mode (consolidate for the entire run ID). This is a configurable flag for wave mode. In the case of stream mode, this flag is defaulted 'No' (discrete mode).
Batch Routing	The batch routing flag dictates whether the routing process is performed in discrete mode (route per order) or in batch mode (route for the whole run ID). This is a configurable flag for wave mode. In the case of stream mode, this flag is defaulted 'No' (discrete mode).
Consolidation Strategy	A consolidation strategy defines different consolidation configurations that drive the consolidation process, such as consolidation criteria.
Cubing Strategy	A cubing strategy defines different configurations that drive the cubing process, such as cubing criteria. You can refer Cubing Process Guide for more details.
Order Planning Mode	The order planning mode defines a mode ('wave', 'stream', 'replenishment) for order planning strategy. Wave mode indicates that the order planning strategy is used for a wave run; stream mode indicates that the order planning strategy is used for a waveless run. Replenishment mode is used for a replenishment wave that creates only replenishment allocations/tasks based on the order need.
Pack Sort Resource Strategy	A pack sort resource strategy drives the resource determination process for 'PACK' or 'PICK/PACK' allocations via the pack sort work criteria. You can refer to the Work Release Functional Guide for more details.
Pick Task Strategy	The pick task strategy drives the task creation process for 'PICK' allocations via the pick task criteria. You can refer to the Work Release Functional Guide for more details.
Pipeline	A pipeline defines the steps (procedures) in a pipeline for order planning strategy and this is defined only for wave mode. In the case of a stream mode, the pipeline is not defined, as stream mode can process orders across multiple pipelines.
Replenishment Resource Strategy	A replenish sort resource strategy drives the resource determination process for 'REPLENISHMENT' allocations via the replenishment resource work criteria. You can refer to the Work Release Functional Guide for more details.
Replenishment Strategy	A replenishment strategy defines different replenishment configurations that drive the replenishment process, such as replenishment inventory criteria and, assignment zone priorities. You can refer Replenishment Process Guide for more details.
Replenishment Task Strategy	A replenish task strategy drives the task creation process for replenishment allocations via replenishment task creation criteria. You can refer to the Work Release Functional Guide for more details.

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Configuration	Description
Routing Strategy	A routing strategy defines different routing configurations that drive the routing process, such as routing criteria. You can refer to the routing process flow for more details (Warehouse Management application).
Selection Criteria	The selection criteria define the selection rules and selection sequencing that drive the order selection process. You can refer DC Order Process or more details.
Strategy Capacity [PPB1]	Capacity can be defined based on the below parameters: • **Maximum orders**: This defines the maximum number of orders that can be selected as part of the specific selection rule. • **Maximum order lines**: This defines the maximum number of order lines that can be selected as part of the specific selection rule. • **Maximum unit**: This defines the maximum number of order quantities in unit terms that can be selected as part of the specific selection rule. • **Maximum weight**: This defines the maximum weight that can be selected as part of the specific selection rule. The maximum weight is calculated based on the order line unit weight and item unit weight and the order quantity. (Weight = Unit Weight * Order Quantity) • **Maximum volume**: This defines the maximum volume that can be selected as part of the specific selection rule. The maximum volume is calculated based on the order line unit volume and item unit volume and the order quantity. (Weight = Unit Volume * Order Quantity)

For details to configure an order planning strategy, you can refer to the DC Order Planning Strategy Configuration Guide.

Selection is limited to no more than 30000 lines.

Order Planning Run Trigger

The order planning strategy screen/page allows the user to initiate an order planning run. An order planning run is a group or a batch of selected orders and order lines, which are processed together according to the order planning mode configured in the order planning strategy. In a wave mode, orders can be processed together at the run level, or discretely on an order-by-order basis. In a streaming mode, the processing is always discretely at the order level, for any orders selected in the order planning run.
Order planning strategy allows two modes of order planning run creation:

- **Time-Triggered Scheduler**: This mode is applicable for both wave and stream order planning runs. In this mode, the scheduler creates a new order planning run at a configured time interval.
- **User-Initiated**: This mode is only applicable for waves. In this mode, you can initiate an order planning run from the order planning strategy screen/page.

Order Planning Run ID Generation

The run ID for a wave and a stream process is generated using next-up number generation logic which is based on the next-up number configuration. There are distinct next-up number configurations for a wave and a stream. Following are the next-up number configurations.

Process	Null
Counter Type	"Stream Run ID" or "Wave Run ID"
Counter Prefix	This identifies the prefix value that is applied to the start of each next-up number.
Start Sequence	Start sequence of the next-up number
End Sequence	End sequence of the next-up number
Current Sequence	Current sequence for the next-up number
Increment By	Incremental value for the next sequence generation
Nextup Length	Length in characters of the next-up number
Append Date Prefix	This is stated as true or false. If true, the system appends the date after the counter prefix. If false, the system does not append any date after the counter prefix.
Date Prefix Format	Displays the different date formats: • YYYYMMDD • MMDDYYYY • DDMMYYYY • YYYYMMDDHHMM • MMDDYYYYHHMM • DDMMYYYYHHMM
Last Generated Date	Date of last next-up number generation
Reset Sequence on Day Change	This is stated as true or false. If true, the sequence is reset to each day at 12 am (midnight).

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Order Pipeline

The pipeline defines the various steps performed during order processing. OPS configuration allows the user to select an existing pipeline to be used for order lifecycle flow. Order planning runs in a wave mode using a single configured pipeline on the OPS, for selected orders having the same pipeline. In a stream mode, an order is processed, DC Order is orchestrated and is based on the associated order pipeline. Therefore, the OPS itself is pipeline-independent and does not contain a pipeline configuration.

For example:

Order Planning Strategy	Mode	Pipeline	Description
Wave_Wholesale	Wave	PL01	Allocation > Cubing > Work Release
Stream_eCom	Stream	N/A	

Note: Selection is not a process defined in the pipeline. However, the selection process is always triggered first before the allocation process step.

Order Planning Run Strategy

Order planning run strategy is a unique instance, which is identified by the order planning run ID and the individual Component Strategy ID, when selected on OPS. When you trigger an order planning run, the DC Order component creates the order planning run strategy, it records and sends the request to components in the pipeline to create a component as a local copy for each component's run-level strategies. Each component validates the run strategy before creating a copy of it. If the validation fails, the order planning run is aborted. DC Order also persists the order planning run strategy locally. These local copies of component-level order planning strategies are used by the components to drive component-level order processing.

Stream_eCom	Stream	NA	Stream_Sel	Stream_Alloc	Stream_Repl	Stream_Cube	Stream_PackSortRes	Stream_Picktask	Stream_Repltask	Stream_ReplRes
Wave_Wholesale	Wave	PL01	Wave_Sel	Wave_Alloc	Wave_Repl	Wave_Cube	Wave_PackSortRes	Wave_Picktask	Wave_Repltask	Wave_ReplRes[PPB1]

When a run is initiated, the following order planning run strategy is created:

Strategy Name	Mode	Pipeline	Run ID	Selection
Wave_Wholesale	Wave	PL01	WAVE001	Wave_Sel

The following selection run strategy is created:

Strategy Name	Run ID	Max Order	Max Lines	Max Units
Wave_Sel	WAVE001	200	0	0

The following allocation run strategy is created:

Strategy Name	Run ID	Facility ID
Wave_Alloc	WAVE001	01
Wave_Repl	WAVE001	01

Pipeline Step Processing

As soon as the run strategies are persisted in each component, DC Order orchestrates the order lifecycle based on the assigned order pipeline. Each step in the pipeline is executed in the sequence configured in the pipeline. A selection and an allocation of an order are performed on the same transactional boundary, and if there is a step present before an allocation (such as routing), it is executed for all the orders. Post-selection happens only when the selected order and the order lines are processed as a part of the planning run. The order planning run is a batch of the selected order lines. In a wave run, a complete order planning run or a batch of selected order lines are processed together or they are discretely based on the order planning strategy configurations. In a streaming run, each selected order is processed discretely and is independent of selected orders.

For details about pipeline processing, you can refer the DC Order Process Guide.

Order Planning Run Status Monitor

The order planning run status monitor job scheduler scans through the order plan run strategy entity to select the run entries in a ‘WAVE’ mode which are currently in a queued status, and this determines if each entry is a candidate for being moved to ‘COMPLETED’ or ‘ABORTED’ status, and if so, it updates accordingly.

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Replenishment Wave

Before running a picking wave, a replenishment wave can be run to create the required replenishments for an order or set of orders so that the required inventory can be kept ready before the picking can happen.
After order selection, the allocation step determines if any replenishment is required at the allocation zones. If yes, only the replenishment allocations are created. Replenishment wave uses only allocation, replenishment, and task creation steps. The pipeline stamped on the order is not used for a replenishment wave.

Stop Wave

Users can stop the wave which is not yet completed and currently running. The system will stop all the further processes which are happening or about to happen in the wave run and will start to unplan the orders and will try to bring the system back to its original state.

- A new action button, ‘Stop Wave' has been introduced in the 'Wave Runs' UI. This action will be available for the wave runs which are not yet completed.
- The DC Order component will broadcast the stop request to the components which are involved in the wave run as per the pipeline configuration.
- Components will honor the stop request and stop processing the requests which are yet to be processed. Components will allow the process to complete which has already started. At the end, the component will send a pipeline response to DC Order. After receiving the pipeline response (success or failure - does not matter here), the DC Order will stop processing further and will trigger the unplanned process for the order.
- When the stop is successful, DCO will update the wave status to ‘Force Cancelled’. The system will also capture the user who initiated the stop along with time stamps in order planning run record.
- As part of the 'Stop Wave' process, the system won't be able to revive the order lines or quantity which got cancelled as part of allocation shortage processing when the stop wave has been registered after the allocation process.

Cancel Unused Replenishment and Unused Bulk Pick during Wave run

- The system has been enhanced to stop exporting the replenishment and pick allocation right after the allocation step in Wave mode.
  - All the allocations will be in the inventory component till the wave is complete and hence wave will have the control to update the allocations or cancel the allocations in case they are unused.
- A new attribute, 'Bulk Distribution Need' is available in allocation so that the system can correctly update this during the allocation step and/or post allocation upon exceptions like order deselection.
- System (inventory component) can cancel the unused replenishment and unused bulk pick.
  - Completely cancel the Replenishment allocation if there is no allocated quantity from the location.
  - Completely cancel the Pick allocation if the pick is completely unused and there is no pack allocation for that item.
  - Partially cancel the Pick allocation when allocation UOM < LPN.

Examples

Example 1 - Unused Replenishment

- Wave is configured to do allocation along with replenishment
- Orders – Multi Line orders

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Order	Line	Item	Quantity
Order 1	Line 1-1	Item A	10
Order 1	Line 1-2	Item B	12
Order 2	Line 2-1	Item A	15
Order 2	Line 2-2	Item B	16

- Item A was allocated after successful replenishment of the reserve zone.
- There was no inventory for Item B in the configured pick zone and replenishment was also not successful in the reserve zones.
- Ultimately, Orders were deselected since Item B could not be allocated and the shortage option was selected as 'Deselect Order'
- Here, the replenishment created for Item A is completely unused and the system will cancel the replenishment allocations within the Wave process.

Example 2 - Complete Unused Bulk Pick

- Wave is configured to do Bulk Allocation with allocation UOM = LPN.

Single line Orders for Item A:

Order	Line	Item	Quantity
Order 1	Line 1-1	Item A	10
Order 2	Line 2-1	Item A	10
Order 3	Line 3-1	Item A	20
Order 4	Line 4-1	Item A	30

Reserve Inventory:

Reserve Location	LPN	Item	Quantity
RESR_01	LPN_01	Item A	70
RESR_01	LPN_02	Item A	70

- After the wave allocation step, LPN_01 was allocated in Bulk mode for these four Orders.
- All four Orders were deselected during the routing step of the wave.
- The pick allocation of 70 units is completely unused.
- Here, the system will cancel this pick allocation within the Wave process.

Example 3 - Partial Unused Bulk Pick

- Wave is configured to do Bulk Allocation with allocation UOM = LPN.

Single line Orders for Item A:

Order	Line	Item	Quantity
Order 1	Line 1-1	Item A	10
Order 2	Line 2-1	Item A	10
Order 3	Line 3-1	Item A	20

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Order	Line	Item	Quantity
Order 4	Line 4-1	Item A	30

Reserve Inventory:

Reserve Location	LPN	Item	Quantity
RESR_01	LPN_01	Item A	70
RESR_01	LPN_02	Item A	70

- After the wave allocation step, LPN_01 was allocated in Bulk mode for the four Orders.
- Let's consider, only Order 3 got deselected during the routing step of the wave.
- Now the pick allocation of 70 Units is partially unused, and only 50 units is needed for these orders.
  - Here, the system will update the 'Bulk Distribution Need' as 50 on the pick allocation.
- Based on the 'Bulk Distribution Need' , the work release component will distribute the 50 units to the available pack allocations and then task creation will happen.
- Users will pull the whole 70 Unit iLPN and pack only 50 units in put into the store area. The user needs to take back the remaining 20 units to a storage location.

Example 4 - Partial Unused Bulk Pick

- Wave is configured to do Bulk Allocation with allocation UOM = Pack

Single line Orders for Item A:

Order	Line	Item	Quantity
Order 1	Line 1-1	Item A	10
Order 2	Line 2-1	Item A	10
Order 3	Line 3-1	Item A	20
Order 4	Line 4-1	Item A	30

Case Pick Inventory:

Case Pick Location	Item	Quantity
CPCK_01	Item A	100 Units (20 Packs)

- After the wave allocation step, the system will allocate 14 Packs in Bulk mode for the four Orders of total need of 70 Units.
- Let's consider, only Order 3 got deselected during the routing step of the wave.
- Now the pick allocation of 14 packs is partially unused and only 10 packs are needed for the Orders need.
  - Here, the system will update the 'Quantity' and 'Bulk Distribution Need' as 50 on the pick allocation.
- Based on the 'Bulk Distribution Need' , the work release component will distribute the 50 units to the available pack allocations and the task creation will happen.
- Users will pick only 10 packs (50 units) and pack them for the orders put in the store area.

Force Pickup Date During Order Planning Strategy Run

Pickup and Delivery Dates can be force changed using the parameters, *Update Pickup and Delivery Dates?*, *New Pickup Start Date time*, and *Days between pickup start date and delivery end date* fields.
When a wave is initiated using an Order Planning Strategy with any option selected from the parameter, *Update Pickup and Delivery*

Dates? the respective dates are updated for orders selected for the wave run based on the option chosen. When the pickup start datetime is updated, the solution automatically adjusts the pickup end date and delivery start date to maintain the correct sequence and prevent routing failures.

Consider the following data before the wave run:

Order	Pickup Start	Pickup End	Delivery start	Delivery End
DO1	13*th* Sep	14*th* Sep	15*th* Sep	16*th* Sep
DO2	13*th* Sep	14*th* Sep	15*th* Sep	20*th* Sep
DO3	17*th* Sep	18*th* Sep	17*th* Sep	18th Sep
DO4	25*th* Sep	26*th* Feb	26*th* Feb	30*th* Sep

Example 1: If ‘Update Pickup and Delivery Dates?’ is selected as ‘Update Pickup Start Date time’ and the pick-up date is selected as 18th Sept

Assume the current date is 17*th* Sept and the pickup date is selected as 18*th* Sept from the *New Pickup Start Date time* field, then the following updates occur after the wave run:

- Pickup start date time is set with the date selected from the ‘New Pickup Start Date time’ field.
- Pickup end date time is set with ‘New Pickup Start Date time’ +1 day
- Delivery start date time is set with ‘New Pickup Start Date time’

Order	Pickup Start	Pickup End	Delivery start	Delivery End
DO1	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	16*th* Sep
DO2	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	20*th* Sep
DO3	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	18th Sep
DO4	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	30*th* Sep

Example 2: If ‘Update Pickup and Delivery Dates?’ is selected as "Update pickup start date time and delivery date times"

Assume the current date is 17*th* Sept, the pickup date is selected as 18*th* Sep from the *New Pickup Start Date time* field, and the ‘Days between the pickup start date and delivery end date’ is specified as 7. The following updates occur after the wave run:

- Pickup start date time is set with the date selected for ‘New Pickup Start Date time’
- Pickup end date time is set with ‘New Pickup Start Date time’ + 24 hours
- Delivery start date time is set with ‘New Pickup Start Date time’
- Delivery end date time is set with ‘New Pickup Start Date time’ + ‘Days between pickup start date and delivery end date’

Order	Pickup Start	Pickup End	Delivery start	Delivery End
DO1	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	**25*th* Sep**
DO2	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	**25*th* Sep**
DO3	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	**25*th* Sep**
DO4	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	**25*th* Sep**

Example 3: If ‘Update Pickup and Delivery Dates?’ is selected as "Adjust past pickup start date time and past delivery date time"

Assume the current date is 17*th* Sept, the pickup date is selected as 18*th* Sep from the *New Pickup Start Date time* field, and the

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‘Days between the pickup start date and delivery end date’ is specified as 7. The following updates occur after the wave run:

- Pickup start date time is set with the date selected for ‘New Pickup Start Date time’
- Pickup end date time is set with ‘New Pickup Start Date time’ + 24 hours
- The delivery start date time is set as ‘New Pickup Start Date time’.
- Similarly, updates the delivery end datetime, if it is less than the current datetime. Delivery end date time is set with ‘New Pickup Start Date time’ plus the number specified in the *Days between pickup start date and delivery end date* field.

Order	Pickup Start	Pickup End	Delivery start	Delivery End
DO1	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	**25*th* Sep**
DO2	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	20*th* Sep
DO3	17*th* Sep	18*th* Sep	17*th* Sep	18th Sep
DO4	25*th* Sep	26*th* Feb	26*th* Feb	30*th* Sep

Example 4: If ‘Update Pickup and Delivery Dates?’ is selected as' Adjust past pickup start date time and update delivery end date time'
Assume the current date is 17*th* Sept, the pickup date is selected as 18*th* Sep from the *New Pickup Start Date time* field, and the ‘Days between the pickup start date and delivery end date’ is specified as 7. The solution updates the pickup start date, pickup end date, and delivery start date only if the pickup start date is less than the current datetime.
The following updates occur after the wave run:

- Delivery end date time is set with ‘New Pickup Start Date time’ + ‘Days between pickup start date and delivery end date’
- Pickup start date time is set with the date selected for ‘New Pickup Start Date time’
- Pickup end date time is set with ‘New Pickup Start Date time’ +1 day
- Delivery start date time is set with the New Pickup Start Date time

Order	Pickup Start	Pickup End	Delivery start	Delivery End
DO1	**18*th* Sep**	**19*th* Sep**	**18*th* Sep**	**25*th* Sep**
DO2	**18*th* Sep**	**19*th* Sep**	**19*th* Sep**	**25*th* Sep**
DO3	17*th* Sep	18*th* Sep	17*th* Sep	**24th Sep**
DO4	25*th* Sep	26*th* Feb	26*th* Feb	**2nd Oct**

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